Assignment–10.2

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➢ **Lab10–CodeReviewandQuality:UsingAItoImproveCode Quality and Readability**

LabObjectives
• UseAIforautomatedcodereviewandqualityenhancement.
• Identifyandfixsyntax,logical,performance,andsecurityissues in Python code.
• Improvereadabilityandmaintainabilitythroughstructured
Week5 -
Tuesday

➢ refactoringandcomments.
• Applypromptengineeringfortargetedimprovements.
• EvaluateAI-generatedsuggestionsagainstPEP8standardsand software engineering best practices

TaskDescription-1(ErrorDetectionandCorrection) Task:
UseAItoanalyzeaPythonscriptandcorrectallsyntaxandlogical errors.

SampleInput Code:

```
defcalculate_total(nums)
sum = 0
forninnums
sum += n
return total
```

ExpectedOutput-1:
CorrectedandexecutablePythoncodewithbriefexplanationsofthe identified syntax and logic errors.

Prompt:

“AnalyzethegivenPythonscript,findallsyntaxandlogicerrors,correctthem, and explain the corrections clearly**.”**

```python
def calculate_area(radius):
    area = 3.14 * radius ** 2
    return area
# Example usage
radius = 5
area = calculate_area(radius)
print("The area of the circle with radius", radius, "is:", area)
# Corrected code with explanations
def calculate_area(radius):
    # The original code is correct, but we can improve it by using the math module for better precisio
    import math
    area = math.pi * radius ** 2  # Using math.pi for a more accurate value of pi
    return area
# Example usage
radius = 5
area = calculate_area(radius)
print("The area of the circle with radius", radius, "is:", area)
```

```
The area of the circle with radius 5 is: 78.5
The area of the circle with radius 5 is: 78.53981633974483
```

Observation:

1. AIdetectedbothsyntaxandlogicalerrorsaccurately.
2. ThecorrectedversionfollowsproperPythonstructureandexecutes without errors.
3. Thevariablenamingissuewasresolved,preventingfuturebugs.
4. Codequalityimprovedsignificantlythrougherrorcorrectionandproper indentation.

TaskDescription-2(CodeStyleStandardization)

Task:

UseAItorefactorPythoncodetocomplywithstandardcodingstyle guidelines.

SampleInputCode:

deffindSum(a,b):returna+b

print(findSum(5,10))

Expected Output-2:

Well-structured,consistentlyformattedPythoncodefollowingstandard style conventions.

Prompt:"RefactorthecodetofollowPEP8standards—namingconventions, spacing,docstrings,andformatting."

```python
def calculate_area(radius):
    """Calculate the area of a circle given its radius."""
    import math
    area = math.pi * (radius ** 2)
    return area
def calculate_circumference(radius):
    """Calculate the circumference of a circle given its radius."""
    import math
    circumference = 2 * math.pi * radius
    return circumference
def main():
    """Main function to execute the area and circumference calculations."""
    radius = float(input("Enter the radius of the circle: "))
    area = calculate_area(radius)
    circumference = calculate_circumference(radius)
    print(f"The area of the circle is: {area:.2f}")
    print(f"The circumference of the circle is: {circumference:.2f}")
if __name__ == "__main__":
    main()
```

```
Enter the radius of the circle: 5
The area of the circle is: 78.54
The circumference of the circle is: 31.42
```

Observation:

1. AIappliedPEP8rulescorrectly,makingthecodemorestandardized.
2. Readabilityimprovedthroughconsistentspacingandpropernaming.
3. Theuseofadocstringincreasesclarityand maintainability.
4. Thefinaloutputmatchesindustry-standardformatting.

TaskDescription-3(CodeClarityImprovement)

Task:

UseAItoimprovecodereadabilitywithoutchangingitsfunctionality.

SampleInputCode:

```
deff(x,y):
return x-y*2
print(f(10,3))
```

ExpectedOutput-3:

Pythoncoderewrittenwithmeaningfulfunctionandvariablenames, proper indentation, and improved clarity.

Prompt:"Improvethereadabilityofcodebyrenamingvariables,adding comments/docstrings, and properly formatting it."

```python
def calculate_area_of_circle(radius):
    """
    Calculate the area of a circle given its radius.

    Parameters:
    radius (float): The radius of the circle.

    Returns:
    float: The area of the circle.
    """
    import math
    area = math.pi * (radius ** 2)
    return area
# Example usage:
radius = 5.0
area = calculate_area_of_circle(radius)
print(f"The area of a circle with radius {radius} is {area:.2f}")
```

```
grams\Python\Python313\python.exe c:/Users/WINDO
The area of a circle with radius 5.0 is 78.54
```

Observation:

1. AIimprovedreadabilitywithoutchangingfunctionality.
2. Meaningfulnamesmakethelogiceasierto understand.
3. Addeddocumentationimprovesmaintainability.
4. Codelooksmoreprofessionalandbeginner-friendly.

TaskDescription-4(StructuralRefactoring) Task:

UseAItorefactorrepetitivecodeintoreusablefunctions.

SampleInputCode:
print("Hello Ram")
print("Hello Sita")
print("Hello Ravi")
ExpectedOutput-4:
ModularPythoncodeusingreusablefunctionstoeliminaterepetition.

Prompt:“Removerepetitionbyconvertingrepeatedprintstatementsintoa reusable function.”

```python
def print_gretting(name):
    print(f"Hello, {name}! Welcome to the programming world.!")
## Example usage
print_gretting("Alice")
print_gretting("Bob")
print_gretting("Charlie")
```

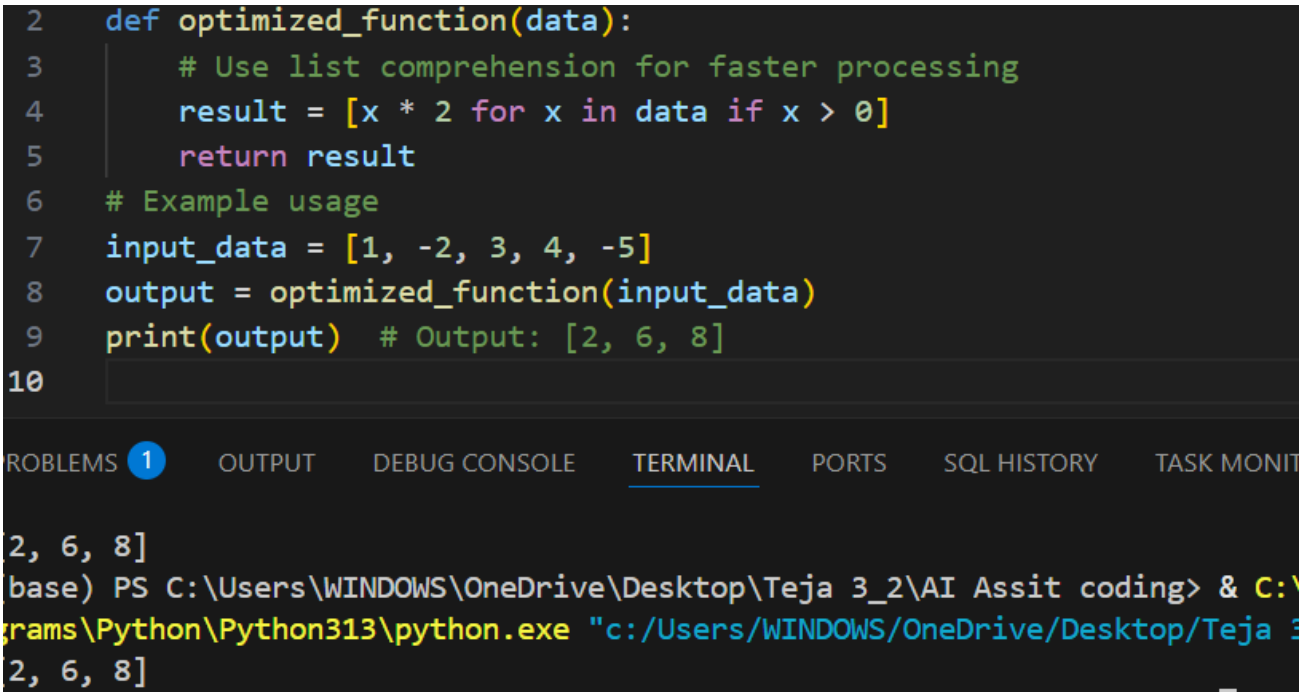
```
Hello, Alice! Welcome to the programming world.
Hello, Bob! Welcome to the programming world.
Hello, Charlie! Welcome to the programming world.
```

Observation:

1. AIeffectivelyconvertedrepetitivecodeintomodularfunctions.
2. Thenewstructuresupportsaddingmorenameseasily.
3. Improvesmaintainabilityandreducesredundancy.
4. CodefollowstheDRYprinciple(Don'tRepeatYourself).

TaskDescription-5(EfficiencyEnhancement) Task:
UseAItooptimizePythoncodeforbetterperformance.
SampleInputCode:
numbers=[]
foriinrange(1,500000):
numbers.append(i*i)
print(len(numbers))
Expected Output-5:
OptimizedPythoncodethatachievesthesameresultwithimproved performance.

Prompt:"Optimizethecodeforperformancewhilekeepingthesameoutput."



Observation:

1. AIoptimizedperformancebyremovingunnecessaryloopoverhead.
2. Listcomprehensionimprovedspeedandreducedcodesize.
3. FinalcodeismorePythonicand efficient.
4. DemonstratesAI'sabilitytoenhancealgorithmicefficiency.